

100 Days of Azure DevOps

Day 58 Handout — A release is a named install of a chart — values.yaml is where environments stop being copy-paste crimes

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — chart vs release vs revision (three different nouns)

Noun	What it is	Command that proves it
Chart	Package: Chart.yaml + templates/ + values.yaml. Not running	helm create charts/myapp ; helm template
Release	Named instance of a chart in a namespace (myapp in workloads)	helm list -n workloads
Revision	Each upgrade increments. Rollback is a revision, not a vibe	helm history myapp -n workloads
values.yaml	Default knobs: image.repository, image.tag, replicaCount	helm upgrade --install ... --set image.tag=...
templates/	Go templates → Kubernetes YAML. Render before you trust it	helm template myapp charts/myapp --set image.tag=demo

Architecture B — one chart, many values files (not charts-dev vs charts-prod)

Constraint	Default	Change when
Env differences	values-dev.yaml / values-lab.yaml overlays	Copying the whole chart per environment — that is the crime
Image tag	--set image.tag=\$(Build.BuildId) from CI	Hardcoded latest in deployment.yaml template
Install vs upgrade	helm upgrade --install (one thought)	helm install that fails the second time because the release exists
No cluster	helm template and read the Deployment it produced	Skipping render because 'the chart looks fine'
Uninstall	helm uninstall myapp -n workloads once in the lab	Abandoned releases as unnamed pets

Architecture C — pitfalls that ship the sample nginx and call it myapp

Pitfall	What you see	Fix
Sample chart untreated	Default nginx image, probes for a different app	helm create is a tour. Delete what is not yours
Tag in the template	CI --set ignored because template does not use .Values.image.tag	templates/deployment.yaml must reference Values
Release vs chart mix-up	helm list empty; charts/ folder full	list shows releases. Package is not an install
kube-system release	helm upgrade --install -n kube-system	-n workloads --create-namespace
Three chart forks	prod missed a securityContext for six months	One chart; values files; reviews on values-prod.yaml

One-liner to remember

Helm is templating for YAML mountains — values.yaml is where environments stop being copy-paste crimes. A chart is a package; a release is what is running.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal repo charts/myapp. Cluster = AKS (if alive) or kind. Render is mandatory even with no cluster.

- 1 helm create charts/myapp. Open Chart.yaml, values.yaml, templates/deployment.yaml. Note image.repository / image.tag. Strip sample nginx assumptions you will not run.
- 2 Point values at <acr>.azurecr.io/myapp. Do not hardcode latest in the template — it must be .Values.image.tag.
- 3 helm template myapp charts/myapp --set image.tag=day58 --namespace workloads | less — confirm Deployment image and namespace. This step counts if you have no cluster.
- 4 If a cluster exists: helm upgrade --install myapp charts/myapp -n workloads --create-namespace --set image.tag=<Build.BuildId> --wait. helm list / helm history.
- 5 Upgrade the tag once more. Confirm revision incremented. Then helm uninstall myapp -n workloads once so you know the opposite of install.
- 6 Write docs/helm-day58.md: chart vs release vs revision in one line each, and the rendered image line. Optional pipeline: HelmDeploy@1 or a script step with the same upgrade --install.

Starter commands — render, then upgrade --install

```
helm create charts/myapp
helm template myapp charts/myapp \
  --namespace workloads \
  --set image.repository=<uniqueacr>.azurecr.io/myapp \
  --set image.tag=$(Build.BuildId)
helm upgrade --install myapp charts/myapp \
  --namespace workloads --create-namespace \
  --set image.repository=<uniqueacr>.azurecr.io/myapp \
  --set image.tag=$(Build.BuildId) --wait
helm list -n workloads
helm history myapp -n workloads
helm uninstall myapp -n workloads
```

Done checklist

- Can distinguish chart, release, and revision out loud
- image.tag comes from values/--set, not a hardcoded latest in the template
- helm template ran; install/upgrade/uninstall once if a cluster exists
- Release namespace is workloads, not kube-system

Tomorrow — Day 59

AKS scaling, monitoring, networking — HPA needs requests, not superstition